SCSE2133 Software Agile Project Management Composition

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Outline

- Introduction to Dynamic Systems Development Method (DSDM)
- The composition of DSDM
- DSDM and Scrum
- Comparison among Scrum, Kanban, and Scrumban
- Introduction to Project Management Tools

Introduction : Dynamic Systems Development Method (DSDM)

Recap…

The DSDM philosophy is that:

"best business value emerges when projects are aligned to clear business goals, deliver frequently and involve the collaboration of motivated and empowered people."

This is achieved when

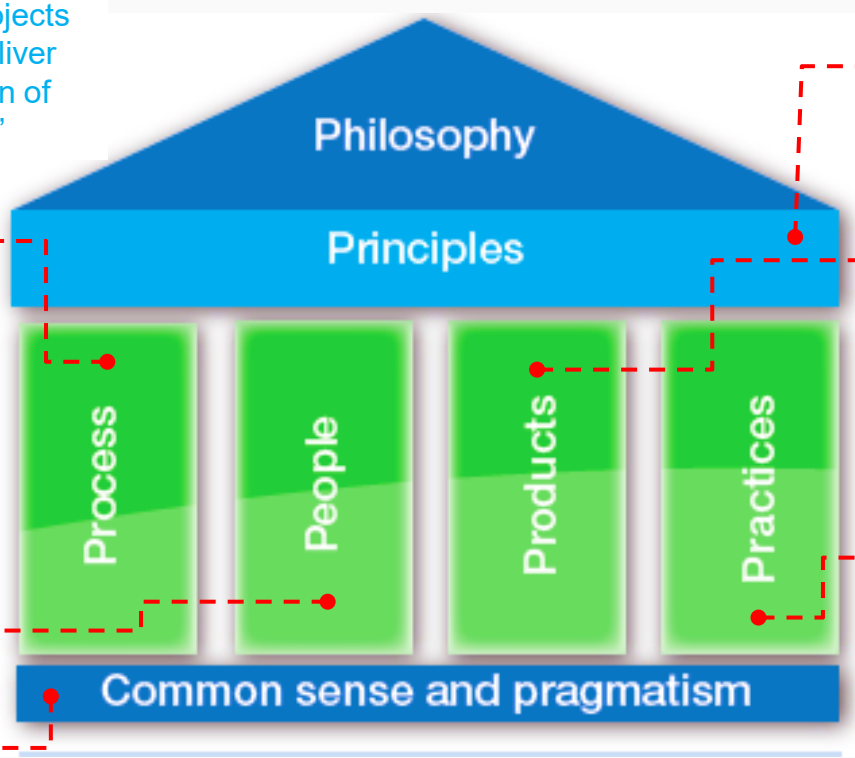
- **Clear Roles & Responsibilities**: DSDM assigns well-defined roles covering business, technical, management, and process interests to ensure balanced representation across the project.

- **Collaboration & Communication**: All team members work closely together, breaking down barriers and fostering effective communication to drive project success.

- **Empowered, Self-Organising Teams**: The best solutions come from teams that are empowered to self-organise and take responsibility within agreed boundaries.

The Composition of DSDM

"best business value emerges when projects are aligned to clear business goals, deliver frequently and involve the collaboration of motivated and empowered people."



- A set of 8 principles that bring the philosophy into practice.

- The lifecycle, phases, timeboxing, iteration, governance flow. Describes how work is structured.

- Roles, responsibilities, team interactions, empowered decision-making. Captures who does what, and how people collaborate.

- The deliverables, artefacts, outputs (business, solution, governance) required. Defines what is produced and tracked.

- Recommended techniques, methods, workshops, modelling, testing, etc. These are tools & approaches to support execution.

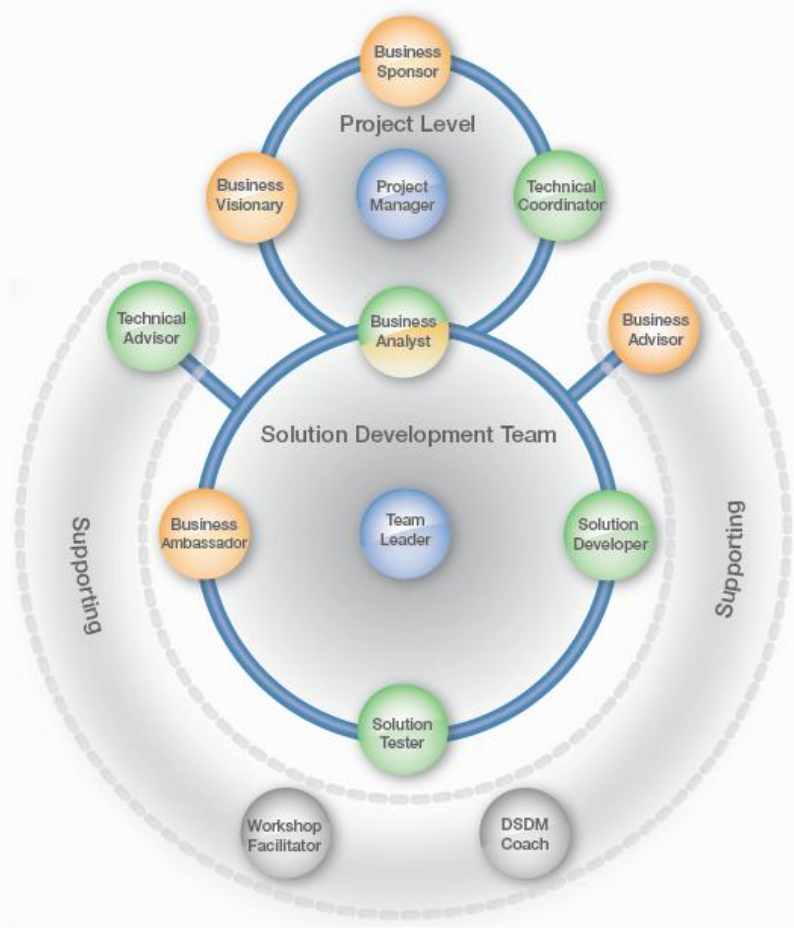
- **Common sense**: using sound practical judgment, independent of over-theory.
- **Pragmatism**: making decisions based on immediate practical consequences rather than rigid theory or dogma.

The People in DSDM

- **Colour Scheme (Roles of Interest):** Orange = Business view, Green = Solution/technical view, Blue = Management/leadership view, **Grey** = Process view, Mixed = Covers multiple views.

- **Role Categories:**
  - **Project-level roles**: Business Sponsor, Business Visionary, Technical Coordinator, Project Manager, Business Analyst → direct, manage, and govern the project.
  - **Solution Development Team roles**: Business Ambassador, Solution Developer, Solution Tester, Business Analyst, Team Leader → engine room of the project, shaping and building the solution.
  - **Supporting roles**: Business Advisors, Technical Advisors, Workshop Facilitator, DSDM Coach → provide guidance, expertise, and support as needed.

- **Role Flexibility:** One person can hold multiple roles, or a role may be shared by several people, provided communication and collaboration remain effective.



DSDM and Scrum

Aspect	DSDM	Scrum
Scope / Lifecycle Coverage	Covers the entire project lifecycle: pre-project, feasibility, foundations, evolutionary development, deployment, post-project.	Primarily focused on the development lifecycle (product backlog → sprints → increments). Governance / project initiation phases are less prescribed.
Governance & Structure	More structured governance, roles, disciplines, and artefacts. Designed to balance agility with control.	Lightweight framework with minimal prescribed governance outside the core roles and events.
Fixed vs Variable Elements	Fixes time, cost, quality early on; scope is variable and managed via MoSCoW prioritisation.	Fixes scope (commitment) per sprint; time is fixed in each sprint, cost is implicit (team capacity).
Requirements / Scope Management	Uses MoSCoW prioritisation to categorise requirements (Must, Should, Could, Won't).	Uses product backlog ordering / prioritisation; changes may be made between sprints but generally backlog is fluid.
Roles & Role Variety	Many roles: Business Sponsor, Business Visionary, Technical Coordinator, Business Analyst, Solution Developer, etc. Rich role structure.	Fewer roles: Product Owner, Scrum Master, Development Team. More lean and focused.
Change Handling	Accepts change, but within boundaries; scope is trimmed if needed (using MoSCoW) to meet time and cost constraints.	Encourages change, backlog is adaptive; within a sprint, changes are limited to preserve focus.
Best Use / Fit	Larger projects needing agility plus governance — especially where project initiation, accountability, stakeholder roles, and full lifecycle governance are needed.	Product development teams wanting agility, fast feedback, minimal overhead, and frequent delivery in iterative cycles.
Complexity & Overhead	Higher overhead because of more roles, more governance, more artefacts and control points.	Lower overhead; lean structure with less prescribed artefacts and roles.

Comparison among Scrum, Kanban, and Scrumban

Aspect	Scrum	Kanban	Scrumban
Description	Lightweight framework to develop and sustain complex products via defined accountabilities, events, and artifacts.	Visual flow method that optimizes work by visualizing tasks and limiting WIP to keep work moving.	Hybrid that combines Scrum's cadence/ceremonies with Kanban's visual flow and WIP limits for more flexibility.
Iterations/Time boxing	Work happens in fixed-length Sprints of one month or less. (Many teams use 1–4 weeks.)	No fixed iterations; teams pull work continuously.	Uses short sprints when helpful plus continuous pull as capacity allows.
Roles & Events	Explicit events: Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective (within each Sprint).	Few mandated roles/ceremonies; teams keep it lightweight and adapt meetings to need.	May keep some Scrum ceremonies (e.g., standups/retros) while applying Kanban flow controls.
Work in Progress (WIP)	WIP is indirectly constrained by the Sprint commitment and Sprint Goal.	Explicit WIP limits per workflow stage to expose bottlenecks and stabilize flow.	Uses Kanban-style WIP limits with simple policies/triggers to manage flow.
Planning & Adaptation	Plan at the start of each Sprint; avoid changes that endanger the Sprint Goal during the Sprint.	Continuous prioritization; new work is pulled as capacity appears.	On-demand planning triggered when the ready backlog drops below a threshold.
Best Use Case	Cross-functional product teams needing structure and a steady delivery cadence.	Ops/support/maintenance or any flow-based work needing adaptability and smooth throughput.	Teams that want a balance of structure and flexibility, or are evolving from Scrum↔Kanban.

Introduction to Project Management Tools

Project Management Tools

- Project management tools are software or systems that support planning, executing, controlling, and closing of projects.
- It help teams organize tasks, monitor progress, allocate resources, communicate, and report status.
- These tools act as enablers — they don't replace good process or governance, but make applying them easier, more visible and efficient.
- The choice of tool depends on factors like project complexity, team size, methodology (Agile, Waterfall, hybrid), and integration needs.
- Here are common categories, with their focus and key features:

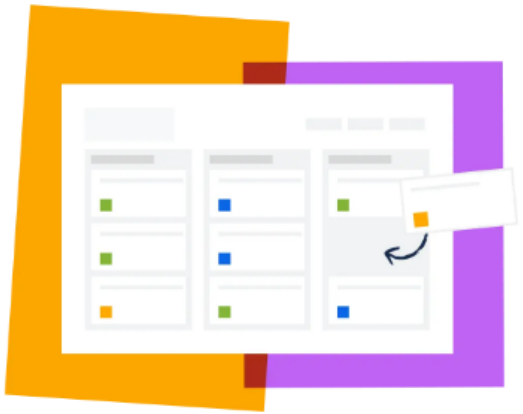
Category	Focus / Purpose	Key Features
All-in-One / Integrated Tools	Provide end-to-end project management coverage	Task planning , scheduling, resource allocation, dashboards, reporting, collaborating.
Task / To-Do Management Tools	Managing individual tasks and simple workflows	Task lists, assignment, due dates, status tracking
Time / Schedule Tools	Focus on timelines and time tracking	Gantt charts, calendars, despendencies, milestones
Resource Management Tools	Allocating and balancing resources	Workload view, availability, conflicts, forecasts
Collaboration / Communication Tools	Enabling team communication and document sharing	Messaging, file sharing, comment threads, versioning
Speacilized / Industry – Specific Tools	Tailored team specific domains or methodologies	For software: backlog + issue tracking for construction: equipment and materials planning: for marketing : campaign timelines
On-Premise vc Cloud / Deployment Types	Based on how the tool is hosted and accessed	Installed servers (on-premise) vs Saas / cloud tools; mobile or desktop clients

Comparison of the Project Management Tools

Tool	Strengths / Key Features	Weaknesses / Trade-offs	Best Fit / Use Case
Jira	Strong support for Agile (Scrum, Kanban ,hybrid) with boards, backlogs, roadmaps, and integrations. Powerful automation templates and workflow customization. Excellent issue / fast tracking, reporting, and visibility across teams.	Can be complex to configure and maintain for non-technical teams. Some features (reporting, advanced planning) may require add-ons or plugins.	Software development teams. Agile teams needing robust issue tracking and process control.
Trello	Simplicity and ease of use ; intuitive Kanban board interface ideal for small teams. Lightweight fast setup, minimal training required	Limited built-in advanced features (reporting, resource management)- relies heavily on integrations of 'Power-Ups'. Less suitable for large or complex projects without customization or add-ons.	Small to medium teams. Simple or visual projects, task tracking.
Clickup	Rich feature set : docs, goals, time tracking , customize workflows, dashboards. Highly customizable with many view options (list board, Gantt, etc.).	Steeper learning curve and complexity in setup. Overhead to maintain configurations and avoid feature overload.	Teams that want an" all-in –one" platform replacing many tools, scaling beyond simple task tracking.

Jira Tool

Project management



Kanban

Manage and track agile work plus integrate developer tools like GitHub



Project management

Designed to help business teams manage work with list and calendar views



Scrum

Plan, prioritize, and schedule sprints using scrum framework

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Summary

Philosophy: The guiding belief — “best business value emerges when projects align with clear business goals, deliver frequently, and involve empowered people.”

Principles: Eight principles serve as the bridge between philosophy and practice, enforcing the mindset and behaviors required to make DSDM effective.

Four Supporting Domains: The philosophy + principles are supported by:

- Process: Lifecycle, phases, timeboxing, governance flows — how work is structured
- People: Roles, responsibilities, collaboration, decision making — who does what
- Products: Deliverables, artefacts, outputs across business, solution, governance — what is produced
- Practices: Techniques, workshops, modelling, testing, other methods — how work is done



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